

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 5: EXPLORING MIXTURES AND THEIR SEPARATION [SET C - NCERT EXEMPLAR SPECIAL]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH05/SET-C****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20):** Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25):** Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 When a mixture of iron filings and sulfur powder is heated strongly in a hard glass test tube, a black mass is formed. When dilute hydrochloric acid is added to this black mass:

- (A) Hydrogen gas (H_2) is evolved which burns with a pop sound
- (B) Hydrogen sulfide gas (H_2S) is evolved which has a foul rotten egg smell
- (C) Sulfur dioxide gas (SO_2) is evolved
- (D) No reaction occurs

Q.2 Why is potash alum (phitkari) added to muddy river water in municipal water treatment plants?

- (A) To kill all bacteria
- (B) The trivalent Al^{3+} ions neutralize the negative surface charges on suspended clay colloidal particles, causing them to coagulate and settle
- (C) To make water sweet
- (D) To increase water boiling point

Q.3 Which of the following can be separated into pure components by fractional distillation?

- (A) A mixture of common salt and sand
- (B) A mixture of petroleum crude oil components (petrol, diesel, kerosene)
- (C) A mixture of oil and water
- (D) A mixture of chalk and water

Q.4 What is the concentration of a saturated solution of potassium chloride in water if 36 g of KCl dissolves in 100 g of water at 20°C ?

- (A) 36.0%
- (B) 26.47%
- (C) 20.0%
- (D) 30.0%

Q.5 Why is an emulsion of oil in water unstable and separates into two distinct layers unless an emulsifier like soap is added?

- (A) Oil is heavier than water
- (B) High interfacial tension between immiscible oil and water drives coalescence; soap molecules bridge both phases to stabilize droplets
- (C) Water evaporates quickly
- (D) Soap changes oil into gas

Q.6 In an experiment, carbon disulfide (CS_2) is added to a test tube containing a mixture of iron filings and yellow sulfur powder. What happens?

- (A) Iron dissolves completely (B) Sulfur dissolves in CS_2 while insoluble iron filings settle at the bottom
(C) Both dissolve completely (D) A violent explosion occurs

Q.7 Which of the following solutions will NOT scatter a beam of light (does not exhibit the Tyndall effect)?

- (A) Starch solution (B) Egg albumin in water
(C) Aqueous copper sulphate (CuSO_4) solution (D) Soap solution

Q.8 The movement of colloidal particles towards the cathode or anode under the influence of an applied external electric field is known as:

- (A) Electrophoresis (Cataphoresis) (B) Dialysis
(C) Electro-osmosis (D) Brownian motion

Q.9 Which method is best suited for recovering both pure water and salt from a seawater sample?

- (A) Simple distillation (B) Filtration through filter paper
(C) Sublimation (D) Centrifugation

Q.10 A student heats ammonium chloride in a china dish covered with an inverted glass funnel whose neck is plugged with cotton. White crystals deposit on:

- (A) The bottom of the china dish (B) The cool upper inner walls of the funnel stem
(C) The outside of the cotton plug (D) No deposit forms

Q.11 Which of the following represents an aerosol colloid where liquid is dispersed in gas?

- (A) Smoke (B) Cloud and fog
(C) Pumice stone (D) Milk of magnesia

Q.12 The solubility of a gas in a liquid solvent typically:

- (A) Increases as temperature increases (B) Decreases as temperature increases
(C) Is unaffected by temperature (D) Is always zero

Q.13 Which technique separates pigments in natural spinach leaf extract into chlorophyll a, chlorophyll b, xanthophyll, and carotene?

- (A) Paper chromatography (B) Fractional distillation
(C) Magnetic separation (D) Centrifugation

Q.14 What is the physical state of dispersed phase and dispersion medium in butter?

- (A) Dispersed phase: Liquid, Dispersion medium: Solid (B) Dispersed phase: Solid, Dispersion medium: Liquid
(C) Dispersed phase: Gas, Dispersion medium: Solid (D) Dispersed phase: Solid, Dispersion medium: Solid

Q.15 Which of the following is an example of an element rather than a compound or mixture?

- (A) Air (B) Water
(C) Pure Diamond (Carbon) (D) Carbon dioxide

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

- (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): Common salt cannot be separated from its water solution by filtration.

Reason (R): Sodium and chloride ions in solution are smaller than 1 nm and easily pass through the pores of filter paper.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.17 Assertion (A): Pure substances possess sharp and fixed melting and boiling points.

Reason (R): In a pure substance, the chemical composition and intermolecular attractive forces are uniform throughout.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.18 Assertion (A): Brownian motion in colloids is caused by gravitational force.

Reason (R): Gravity pulls colloidal particles downwards towards the bottom of the container.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.19 Assertion (A): Water is considered a chemical compound rather than a mixture of hydrogen and oxygen gases.

Reason (R): Water has completely different properties from its constituent elements and its components cannot be separated by physical means.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.20 Assertion (A): Adding common salt to pure water increases its boiling point and decreases its freezing point.

Reason (R): Salt is an ionic electrolyte that dissociates into Na^+ and Cl^- ions, altering the solvent's vapor pressure.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Chemical Differentiation between Mixture and Compound of Iron and Sulfur

Two chemistry students, Aman and Neha, are given identical portions containing 7 g of iron filings and 4 g of sulfur powder.

- **Student Aman:** Crushes and mixes the iron and sulfur thoroughly at room temperature in a mortar (Sample 1).
- **Student Neha:** Heats the mixture strongly in a crucible until it glows with a red flame and forms a hard, black fused mass (Sample 2).

Both students test their samples with a bar magnet, carbon disulfide solvent (CS_2), and dilute hydrochloric acid (HCl).

Q.21 When a bar magnet is brought near Sample 1 (Aman's sample):

- (A) Nothing is attracted (B) Iron filings are attracted to the magnet leaving yellow sulfur powder behind
 (C) Both iron and sulfur stick to the magnet (D) The magnet loses its magnetism

Q.22 When Neha tests Sample 2 with the bar magnet, what is observed?

- (A) Iron filings are pulled out easily
(B) The entire black mass is non-magnetic / does not separate into iron and sulfur
(C) The sample explodes
(D) Sulfur turns blue

Q.23 When carbon disulfide (\$CS_2\$) is added to Sample 1, what happens?

- (A) Iron dissolves
(B) Sulfur dissolves, and on evaporating the liquid, yellow sulfur crystals are recovered
(C) Both substances remain insoluble
(D) The beaker melts

Q.24 When dilute hydrochloric acid is added to Aman's Sample 1, what gas is evolved?

- (A) Hydrogen gas (\$H_2\$) which burns with a pop sound
(B) Rotten egg smelling hydrogen sulfide (\$H_2S\$)
(C) Chlorine gas
(D) Oxygen gas

Q.25 Which general principle of chemistry is demonstrated by Neha's Sample 2 forming a compound?

- (A) Law of conservation of volume
(B) A compound has entirely different chemical and physical properties from its constituent elements
(C) Compounds are always gases
(D) Heat destroys all mass

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(B)	Heating chemically forms iron(II) sulfide ($\text{Fe} + \text{S} \rightarrow \text{FeS}$). Adding dilute acid produces toxic hydrogen sulfide gas ($\text{FeS} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2\text{S} \uparrow$) with a characteristic rotten egg odor.
Q.2	(B)	According to the Hardy-Schulze rule, multivalent counter-ions (Al^{3+}) neutralize the electric charge of colloidal clay particles, causing rapid flocculation and sedimentation.
Q.3	(B)	Petroleum is a complex mixture of miscible hydrocarbons with differing boiling points separated by industrial fractional distillation columns.
Q.4	(B)	$\text{Mass of solution} = 36 + 100 = 136 \text{ g}$. $\text{Percentage} = \frac{36}{136} \times 100 \approx 26.47\%$.
Q.5	(B)	Soap is an amphiphilic emulsifier: its hydrophobic tail dissolves in oil droplets while its hydrophilic head interacts with water, forming micelles that prevent droplet coalescence.
Q.6	(B)	Sulfur is soluble in organic carbon disulfide (CS_2), whereas iron is insoluble, allowing physical separation of the mixture.
Q.7	(C)	Copper sulphate forms a true homogeneous solution where solute ions are $< 1 \text{ nm}$ in diameter, too small to scatter visible light rays.
Q.8	(A)	Electrophoresis is the migration of electrically charged colloidal particles toward the oppositely charged electrode in an electric field.
Q.9	(A)	Simple distillation vaporizes water, which condenses into pure liquid in the receiver, leaving solid salt behind in the distillation flask.
Q.10	(B)	Ammonium chloride sublimates upon heating; its vapors rise and cool on the colder funnel walls, forming solid sublimate crystals.
Q.11	(B)	Clouds and fog consist of liquid water droplets dispersed in atmospheric air (gas).
Q.12	(B)	Dissolution of gases in liquids is exothermic; increasing temperature drives dissolved gas molecules out into the vapor phase (e.g., boiling water drives out dissolved air).
Q.13	(A)	Paper chromatography separates plant pigments based on differential partitioning between cellulose paper and a moving solvent (petroleum ether and acetone).
Q.14	(A)	Butter is a water-in-oil emulsion / gel where liquid water droplets are dispersed throughout a solid fat matrix.
Q.15	(C)	Diamond is an allotropic form of pure elemental carbon (C).
Q.16	(A)	True solution particles are dissociated ions of sub-nanometer scale, passing unobstructed through cellulose filter pores.
Q.17	(A)	Because all particles in a pure chemical species are identical, phase transitions occur at precise characteristic temperatures, unlike mixtures which melt over a range.
Q.18	(D)	Assertion is false: Brownian motion is caused by continuous unbalanced thermal impacts of dispersion medium molecules, not gravity. Reason is a true physical fact.
Q.19	(A)	Water (H_2O) is chemically bonded in a fixed 1:8 mass ratio, exhibits totally distinct chemical properties (extinguishes fire while H_2 burns and O_2 supports combustion), and requires electrolysis to split.
Q.20	(A)	Dissolved non-volatile solutes lower vapor pressure, causing boiling point elevation and freezing point depression (colligative properties).
Q.21	(B)	Sample 1 is a physical mixture; iron retains its magnetic properties and is separated cleanly by a magnet.
Q.22	(B)	Sample 2 is a chemical compound (FeS). Chemical combination alters the magnetic property; iron atoms are chemically bonded and cannot be separated physically.
Q.23	(B)	Free sulfur dissolves in organic CS_2 . Evaporating the filtrate recovers rhombic sulfur crystals.

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.24	(A)	In Sample 1, free iron reacts with acid ($\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2 \uparrow$), evolving odorless, combustible hydrogen gas.
Q.25	(B)	Chemical synthesis forms new bonds, producing compounds with properties entirely distinct from the constituent elements.